

APPLICATION FORMAT FOR THE IDEATION STAGE FOR ROBOT MAKING TECHNICAL DETAILS FOR PROPOSED ROBOT

PROPOSED ROBOT — AUTONOMOUS SOIL REMEDIATION ROBOT

Robots for Climate Change Challenges

Item	Particular	Proposed Details
1	Level of Robot Making	Senior
2	Category of Robot	Application Based Robot (ABR-I) — Robots for Climate Change Challenges
3	Primary Function	Autonomous detection, selective collection and closed-loop verification of simulated soil-pollution patches.

(Wherever necessary separate sheet/page is allowed to attach; Institute may submit extra details if find necessary)

Additional CAD Design File

Proposed Robot Assembly Design

The complete proposed robot assembly has been designed in Autodesk Fusion 360. The Fusion 360 CAD file is attached separately as an additional supporting file for detailed reference and evaluation of the robot's mechanical design and component arrangement.

Attachment: Proposed Robot Assembly Design – Fusion 360 CAD File

3. Robot Assembly Design (Proposed Diagram)

The proposed robot is a compact four-wheel mobile ground vehicle with a rigid JCB-style collection arm. Electrically actuated joints position the collector, while a near-collector sensing assembly measures the selected soil parameter during the collection cycle. A removable upper container stores collected simulated polluted soil.

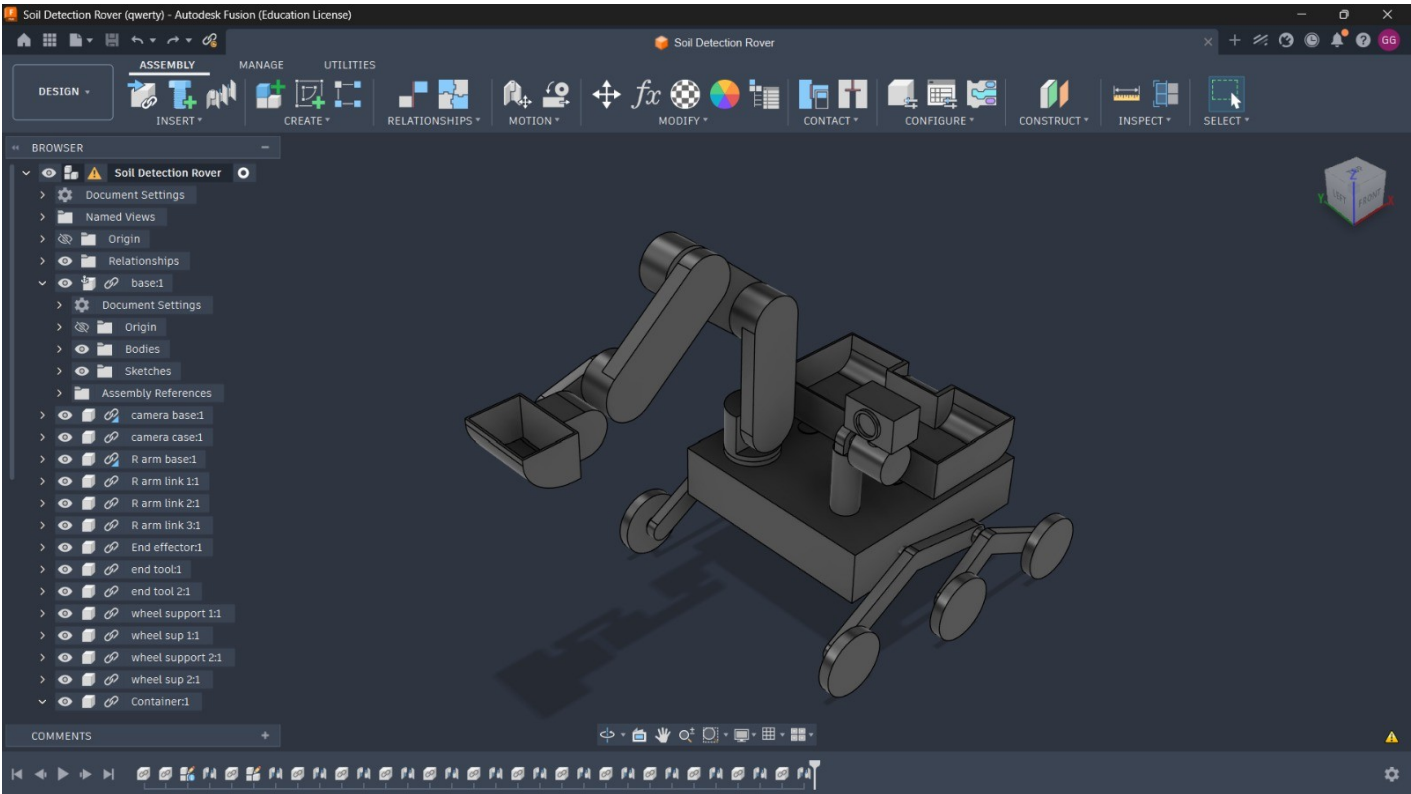


Figure 1. Proposed overall robot assembly — CAD concept/reference render.

Recommended CAD sheets for the final submission:

- Isometric / overall assembly view
- Front, side and top views with final dimensions
- Arm and joint assembly detail
- Collector/scoop with sensor mounting detail
- Exploded assembly view

Subsystem	Design choice	Purpose
Mobile base	4-wheel drive chassis	Stable movement and controlled positioning
Collection mechanism	Multi-joint rigid arm + scoop	Dig, lift and transfer target soil
Sensing	Camera + obstacle sensor + soil sensor	Target detection, safe motion and post-collection verification
Storage	Removable upper container	Temporary storage of collected material
Control	Raspberry Pi + ESP32	High-level decision making with real-time actuator control

4. Components to be Used

The following components are proposed for the proof-of-concept robot. Components are grouped according to the prescribed format. Photographs are intentionally kept outside the tables so that the procurement summary remains compact and the technical justification can be read alongside each photograph. Approximate prices are indicative and should be rechecked on ROBU.in before purchase.

a. List of Structure Components

Sr. No.	Component	Specification	Make / Company	Approx. Price (Rs.)
1	Aluminium T-slot profile	20 mm × 20 mm	Generic / ROBU.in	₹200–400 / m
2	Aluminium plate	≈2 mm thick	Generic / ROBU.in	₹250–400 / pc
3	Ball bearings	8 × 22 × 7 mm class	Generic / ROBU.in	₹40–80 / pc
4	Fasteners / T-nuts	M3/M4 bolts, nuts, T-nuts	Generic / ROBU.in	₹150–250 / set
5	Drive wheels	≈100 mm class	ROBU.in	₹108–224 / pc

b. List of Motion Components

Sr. No.	Component	Specification	Make / Company	Approx. Price (Rs.)
1	12 V geared DC motor	≈100 RPM, encoder- compatible	ROBU.in	₹589–849 / pc
2	High-torque servo	MG996R / metal gear	ROBU.in	₹350–450 / pc
3	High-torque arm servo	20 kg-class metal gear	ROBU.in	₹2,100–2,500 / pc
4	GT2 timing belt	GT2 class	Generic / ROBU.in	₹150–250 / m
5	Pulley / shaft coupler	GT2 / shaft coupling	Generic / ROBU.in	₹80–150 / pc

c. List of Electronics Components

Sr. No.	Component	Specification	Make / Company	Approx. Price (Rs.)
1	ESP32 controller	Wi-Fi/Bluetooth MCU; low-level control	ROBU.in	₹350–900 / pc
2	Raspberry Pi 4 Model B	4 GB class; high- level processing	ROBU.in	₹5,600–14,500 / pc*
3	L298N motor driver	Dual H-bridge; drive motors	ROBU.in	₹100–150 / pc
4	PCA9685 servo driver	16-channel, 12-bit PWM	ROBU.in	₹219–369 / pc
5	Camera module	Raspberry Pi- compatible camera	ROBU.in	₹900–2,000 / pc
6	Ultrasonic sensor	HC-SR04 class	ROBU.in	₹69–220 / pc
7	Soil sensor	Capacitive moisture / selected soil parameter	ROBU.in	₹8000–9000 / pc
8	12 V rechargeable battery	Suitable current rating for drive system	ROBU.in / Generic	₹900–1,800 / pack

d. List of Other Accessories

Sr. No.	Component	Specification	Make / Company	Approx. Price (Rs.)
1	Wire & connectors	Jumper wires, XT60 / terminal connectors	Generic / ROBU.in	₹200–350 / set
2	Cable sleeves / heat-shrink	Electrical insulation and protection	Generic / ROBU.in	₹80–200 / set
3	Cable ties	Nylon, assorted sizes	Generic / ROBU.in	₹50–100 / set

Details on Structural Components:

1. Aluminium Extrusions (2020)



Purpose: To provide a strong, lightweight and modular structural frame for the chassis and arm supports.

Features:

- 20 mm × 20 mm T-slot geometry provides convenient attachment points.
- Good strength-to-weight ratio for a compact mobile platform.
- Bolted construction allows components to be repositioned without welding.

Second Order Impacts: The modular frame reduces fabrication time and makes later CAD-driven changes easier to implement.

Why this component? The T-slot profile combines structural stiffness with rapid assembly and reconfiguration.

2. Aluminium Plates



Purpose: To form mounting panels, protective covers and flat interfaces for electronic and mechanical assemblies.

Features:

- Low mass compared with steel plates of similar size.
- Can be drilled, cut and bolted to the T-slot frame.
- Provides a flat surface for controllers, drivers and brackets.

Second Order Impacts: Replaceable panels keep sensitive electronics separated from the structural frame and simplify servicing.

Why this component? Aluminium plate is practical for lightweight enclosure and mounting work while remaining compatible with the modular frame.

Details on Structural Components:

3. Ball Bearings



Purpose: To support rotating shafts and reduce friction at wheel, arm and linkage joints.

Features:

- Reduces sliding friction and shaft wear.
- Provides smoother rotation under load.
- Supports repeatable movement of mechanical joints.

Second Order Impacts: Proper bearing support reduces mechanical losses and helps maintain alignment during repeated operation.

Why this component? Bearings are used wherever a rotating shaft needs stable support rather than relying on the structural material directly.

4. Fasteners and T-Nuts



Photograph / reference image: Fasteners and T-Nuts

Purpose: To assemble the T-slot frame, mounting plates, brackets and serviceable modules without permanent joints.

Features:

- M3/M4 fasteners are suitable for compact robotic assemblies.
- T-nuts allow components to be mounted directly into T-slots.
- Bolted joints permit maintenance and replacement.

Second Order Impacts: Serviceable joints reduce repair time and allow the robot to evolve as the prototype is tested.

Why this component? Fasteners preserve the modular nature of the chassis and avoid welding-dependent fabrication.

5. Drive Wheels



Purpose: To convert geared motor output into controlled ground motion while maintaining contact with the test surface.

Features:

- Approximately 100 mm wheel class suits a compact ground robot.
- Provides clearance over small surface irregularities.
- Works with geared DC motors for controlled low-speed movement.

Second Order Impacts: Wheel diameter affects traction, ground clearance, turning effort and motor load.

Why this component? The selected wheel size provides a practical compromise between compactness and terrain handling for the proof-of-concept platform.

Details on Motion Components:

6. 12 V Geared DC Motor



Photograph / reference image: 12 V Geared DC Motor

Purpose: To provide the main propulsion torque for the mobile base.

Features:

- Approximately 100 RPM output is suitable for controlled robotic motion.
- Gear reduction increases available torque at the wheel.
- Encoder compatibility can support closed-loop odometry when an encoder is fitted.

Second Order Impacts: Closed-loop wheel feedback can improve repeatability of approach and positioning near the target collection zone.

Why this component? A geared DC motor provides the torque needed for a loaded chassis while retaining simple PWM-based speed control.

7. High-Torque Servo (MG996R Class)



Purpose: To actuate compact joints where controlled angular positioning is required.

Features:

- Metal-gear construction is suited to repeated joint motion.
- Position control allows arm geometry to be commanded directly.
- Compact packaging helps keep the arm lightweight.

Second Order Impacts: Servo-based joints simplify calibration because commanded joint position can be mapped to a known arm pose.

Why this component? The MG996R class provides a practical balance of torque, size and cost for prototype joints. Exact purchased model should be photographed after procurement.

Details on Motion Components:

8. High-Torque Arm Servo (20 kg-Class)



Purpose: To drive the higher-load arm joint where the collector must be lifted or positioned against soil resistance.

Features:

- Higher torque capacity than a small hobby servo.
- Metal gearing supports repeated load cycles.
- Provides controlled angular movement for the main lifting joint.

Second Order Impacts: Higher joint torque reduces the likelihood of stall during lifting and allows the collector to be positioned more reliably.

Why this component? The high-load joint is the most demanding part of the arm, so a higher-torque servo is used instead of a small standard servo. Exact purchased model should be photographed after procurement.

9. GT2 Timing Belt



Photograph / reference image: GT2 Timing Belt

Purpose: To transmit rotary motion between a motor and a remote shaft where direct coupling is inconvenient.

Features:

- Toothed engagement reduces slip compared with a smooth belt.
- Flexible routing helps separate the motor from the joint.
- Useful where compact transmission is required.

Second Order Impacts: Remote transmission can improve packaging and reduce the mass placed directly on a moving joint.

Why this component? A timing belt is selected when the design requires synchronized rotary transmission without a large gear train.

10. Pulley / Shaft Coupler



Photograph / reference image: Pulley / Shaft Coupler

Purpose: To connect the motor or belt drive to the corresponding shaft while maintaining reliable torque transfer.

Features:

- Provides a defined mechanical interface between shafts.
- Allows alignment and service access.
- Works with the selected belt or direct-drive arrangement.

Second Order Impacts: A serviceable coupling reduces the need to redesign the complete actuator assembly if a motor or shaft is replaced.

Why this component? The coupling is the mechanical interface between standard motor shafts and the custom arm mechanism.

Details on Electronics and Control Components:

11. ESP32 Controller



Photograph / reference image: ESP32 Controller

Purpose: To execute low-level real-time control, actuator safety checks and communication with the high-level computer.

Features:

- Provides PWM-capable outputs for motor and servo control.
- Compact controller suitable for embedded mounting.
- Wi-Fi/Bluetooth capability supports development and diagnostics.

Second Order Impacts: Separating low-level control from high-level processing makes the system more robust because actuator commands remain under a dedicated controller.

Why this component? ESP32 is used as the low-level control layer so the high-level computer does not directly drive motors and servos.

12. Raspberry Pi 4 Model B



Photograph / reference image: Raspberry Pi 4 Model B

Purpose: To run high-level perception, decision-making and robot coordination functions.

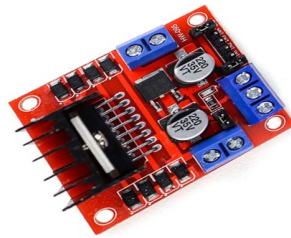
Features:

- Provides substantially more computing capability than a small MCU.
- Supports camera-based processing and robotics software.
- Acts as the high-level decision layer above the ESP32.

Second Order Impacts: High-level processing can be upgraded in software without changing the low-level actuator interface.

Why this component? The Raspberry Pi provides computing resources for vision and autonomous task logic while the ESP32 handles time-critical outputs.

13. L298N Motor Driver



Photograph / reference image: L298N Motor Driver

Purpose: To interface the low-level controller with the DC drive motors through an H-bridge switching stage.

Features:

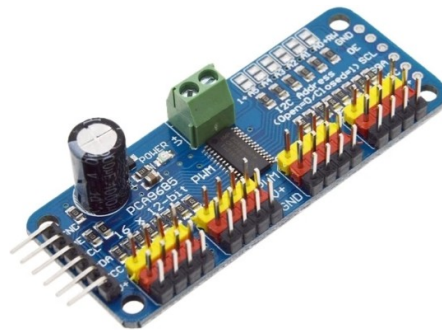
- Dual-channel motor control.
- Supports forward/reverse direction and PWM speed control.
- Provides a simple prototype interface.

Second Order Impacts: A separate driver prevents the controller from directly handling motor current and simplifies debugging.

Why this component? The driver is placed between the ESP32 and drive motors so the controller supplies only logic-level commands.

Details on Electronics and Control Components:

14. PCA9685 Servo Driver



Photograph / reference image: PCA9685 Servo Driver

Purpose: To generate multiple stable PWM channels for the arm servos while reducing timing-sensitive outputs required from the controller.

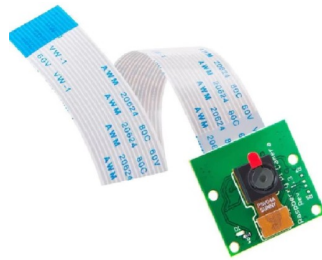
Features:

- Provides multiple PWM channels for several servos.
- Useful for coordinated multi-joint arm control.
- Communicates with the controller through a digital interface.

Second Order Impacts: Dedicated PWM generation improves scalability when additional joints or auxiliary servos are added.

Why this component? A dedicated servo driver keeps multi-joint actuation organized and reduces software complexity on the main controller.

15. Camera Module



Photograph / reference image: Camera Module

Purpose: To observe the test arena and provide visual information for identifying the defined target region.

Features:

- Compact and lightweight for mounting on the robot.
- Compatible with the Raspberry Pi processing layer.
- Can provide image data for target localization.

Second Order Impacts: Vision allows the robot to use the same platform for perception and task execution instead of relying only on fixed external markers.

Why this component? The camera is the primary visual input for high-level target detection and localization.

16. Ultrasonic Distance Sensor



Photograph / reference image: Ultrasonic Distance Sensor

Purpose: To provide short-range obstacle and clearance information during mobile-base movement.

Features:

- Non-contact distance measurement.
- Low-cost and simple interface.
- Useful as an additional safety input near obstacles.

Second Order Impacts: Distance feedback can trigger a stop or motion adjustment before the chassis reaches an obstacle.

Why this component? The sensor provides a simple independent safety layer for short-range obstacle detection.

Details on Electronics and Control Components:

17. Soil Sensor



Photograph / reference image: Soil Sensor

Purpose: To measure the selected soil parameter near the collector and provide feedback for the post-collection verification step.

Features:

- Placed close to the collection region for localized measurement.
- Provides an electrical signal that can be sampled by the control system.
- Can support comparison before and after a collection pass.

Second Order Impacts: Post-collection sensing creates the closed-loop behaviour: the robot can decide whether another collection pass is required.

Why this component? The soil sensor is central to the remediation concept because it links physical collection to measurable verification.

18. 12 V Rechargeable Battery Pack



Photograph / reference image: 12 V Rechargeable Battery Pack

Purpose: To provide the primary electrical energy source for the mobile base and associated electronics.

Features:

- Supplies the main DC bus for the drive system.
- Can be combined with regulated rails for logic electronics.
- Mounted securely to maintain the robot centre of gravity.

Second Order Impacts: A dedicated energy source allows untethered operation during the demonstration and simplifies field setup.

Why this component? A rechargeable 12 V pack matches the proposed drive-motor voltage while allowing regulated power rails for lower-voltage electronics.

Details on Other Accessories:

19. Wire and Connectors



Purpose: To provide reliable electrical connections between the battery, drivers, controllers, sensors and actuators.

Features:

- Organized connectors simplify assembly and maintenance.
- Power connectors can be selected according to expected current.
- Signal wiring can be separated from high-current motor wiring.

Second Order Impacts: Good wiring practice reduces intermittent faults and makes troubleshooting faster during testing.

Why this component? Connectors are used at service points so modules can be replaced easily.

20. Cable Sleeves / Heat-Shrink and Cable Ties



Purpose: To protect, route and secure the electrical harness along the robot frame and moving arm.

Features:

- Protects wires from abrasion and accidental pulling.
- Keeps cables away from wheels and moving joints.
- Improves serviceability and overall assembly quality.

Second Order Impacts: A controlled cable path reduces the probability of cable damage becoming a mechanical or electrical failure.

Why this component? These low-cost accessories are important for safe routing around moving mechanisms.

5. Methodology of Making Robot

The robot is designed as a closed-loop environmental monitoring and collection platform. During the demonstration, a defined simulated soil-pollution patch is detected, localized, approached, collected, and checked again. The post-collection measurement determines whether another collection pass is required.

Overall sequence: SCAN → DETECT → LOCALIZE → NAVIGATE → POSITION ARM → COLLECT → VERIFY → DECIDE → RE-COLLECT / COMPLETE → STORE → NEXT TARGET

5.1. System Flowchart

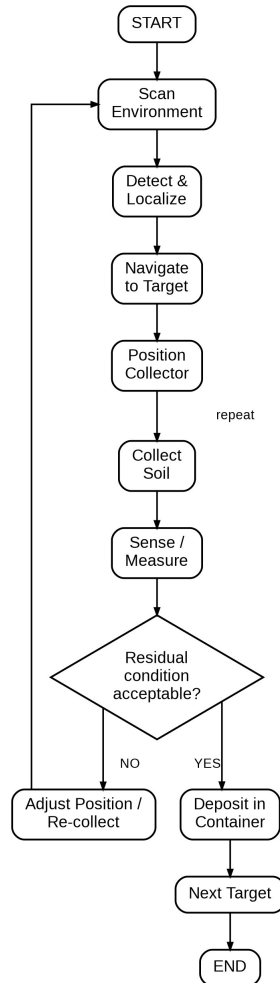


Figure 2. Closed-loop operating sequence. The flowchart is intentionally black-and-white for technical-document clarity.

5.2. Key Operations

Detection & localization: The camera scans the arena and identifies the defined simulated pollution patch and its approximate location.

Navigation: The mobile base moves toward the target using the high-level controller and obstacle/distance sensing.

Collection: The arm positions the scoop, collects the target material and lifts it toward the storage container.

Verification: The soil-sensing assembly measures the selected parameter again after collection.

Decision: If residual pollution remains above the selected criterion, the controller adjusts the collection position and repeats the collection pass.

Storage & task update: If the target satisfies the criterion, the material is deposited in the container and the robot proceeds to the next target.

5.3.JCB-Style Arm and Collector

The rigid arm uses multiple electrically actuated joints. The base joint positions the arm, the shoulder and elbow control reach and lift, and the collector joint controls scoop orientation. The scoop is designed as a replaceable part so the collector geometry can be modified during prototyping.

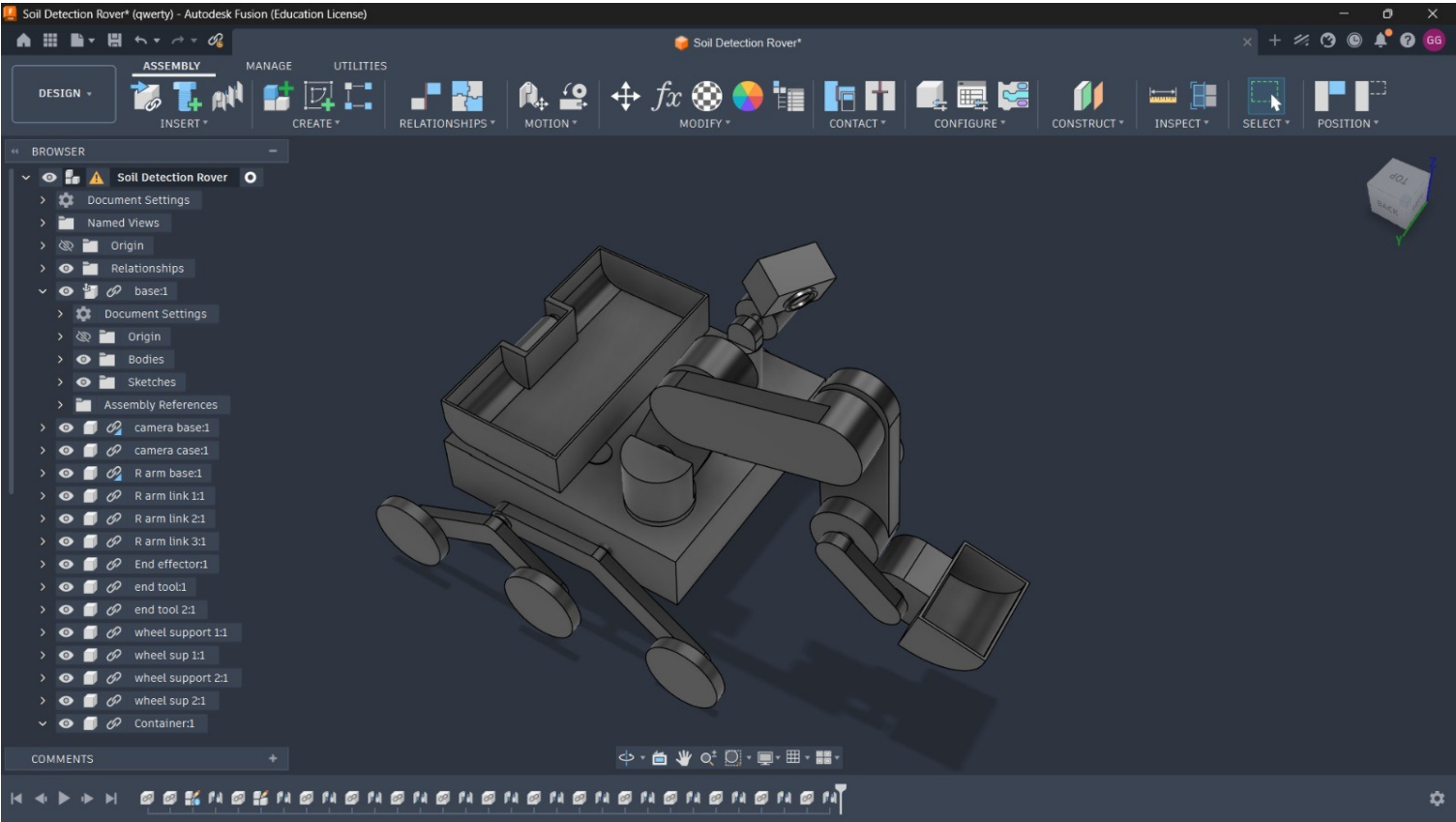


Figure 3. Arm, scoop and collection-container concept.

5.4. Closed-Loop Verification

After a collection pass, the robot re-senses the target region. The controller compares the measured/derived residual condition against the selected remediation criterion. Residual pollution causes a re-collection command; a satisfactory result causes the target to be marked complete.

Collect Soil	Measure	Criterion Check	YES	NO
Collection pass	Post-collection sensing	Residual condition	Deposit / Complete	Adjust position / Re-collect

Verification logic: the post-collection measurement directly affects the next action; this is the main closed-loop behaviour of the proposed system.

5.5. Electronics & Control Architecture

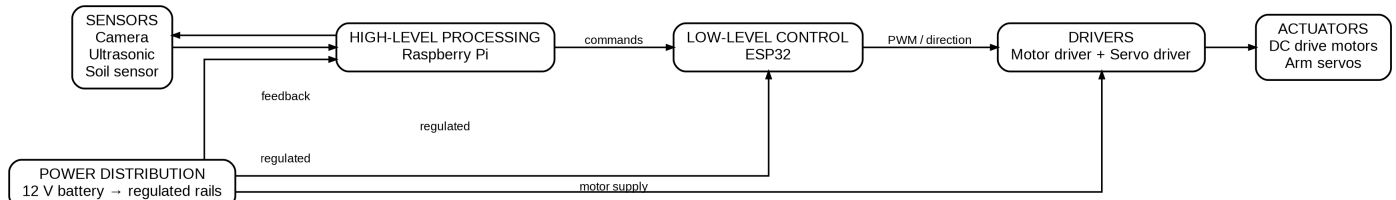


Figure 4. Sensors → high-level processing → low-level control → drivers → actuators, with feedback and regulated power distribution.

5.6. Safety and Failure Handling

- Low-level control limits actuator commands to the defined mechanical travel and servo limits.
- Obstacle sensing causes the mobile base to stop or alter the motion command when an obstacle is inside the defined safety range.
- An emergency power-stop is provided in the electrical architecture.
- Motor/servo power is separated from logic power through regulated distribution to reduce controller resets.
- Collection is inhibited if the arm is outside its safe operating envelope.

5.7. Proposed Technical / Novel Integration

Feature	Proposed implementation
Localized target detection	Vision-based identification of defined simulated pollution patches.
Selective collection	JCB-style scoop is positioned over the detected target rather than collecting the entire area.
Near-collector sensing	Soil-sensing assembly is placed at/near the collector for localized measurement.
Closed-loop verification	Post-collection sensing determines whether another pass is required.
Adaptive re-collection	Residual pollution changes the next collection position/trajectory.
Task update	A completed target is recorded before selecting the next target.
User feedback	Pollution status and operating state can be shown on a display.

6. Application of Proposed Robot in a Societal Context

Autonomous detection and cleanup of localized soil pollution.

7. Size of Robot Proposed for Proof of Concept (Small Version)

Dimension	Proposed Value
Length	28 cm
Width	28 cm
Height	29 cm

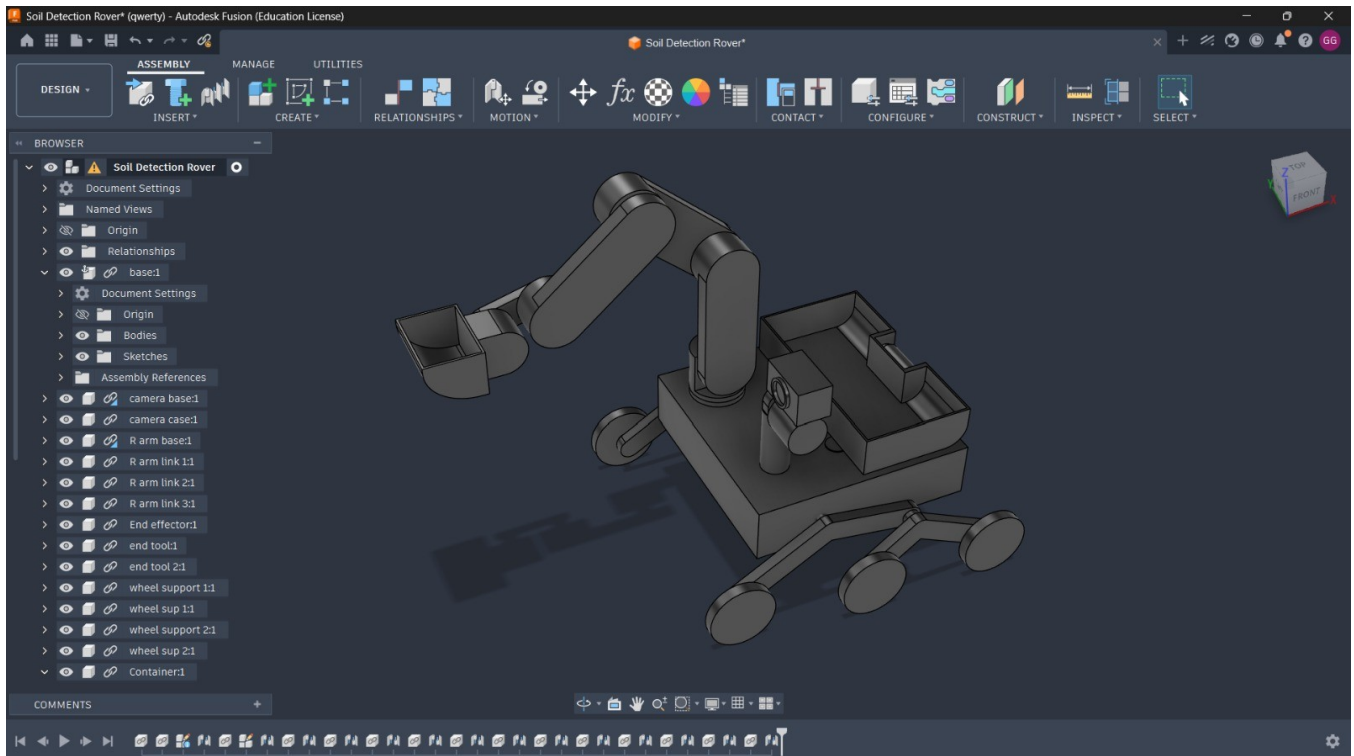
8. Size of Robot Proposed as Prototype (Actual Version)

Dimension	Final CAD Value
Length	29 cm
Width	29 cm

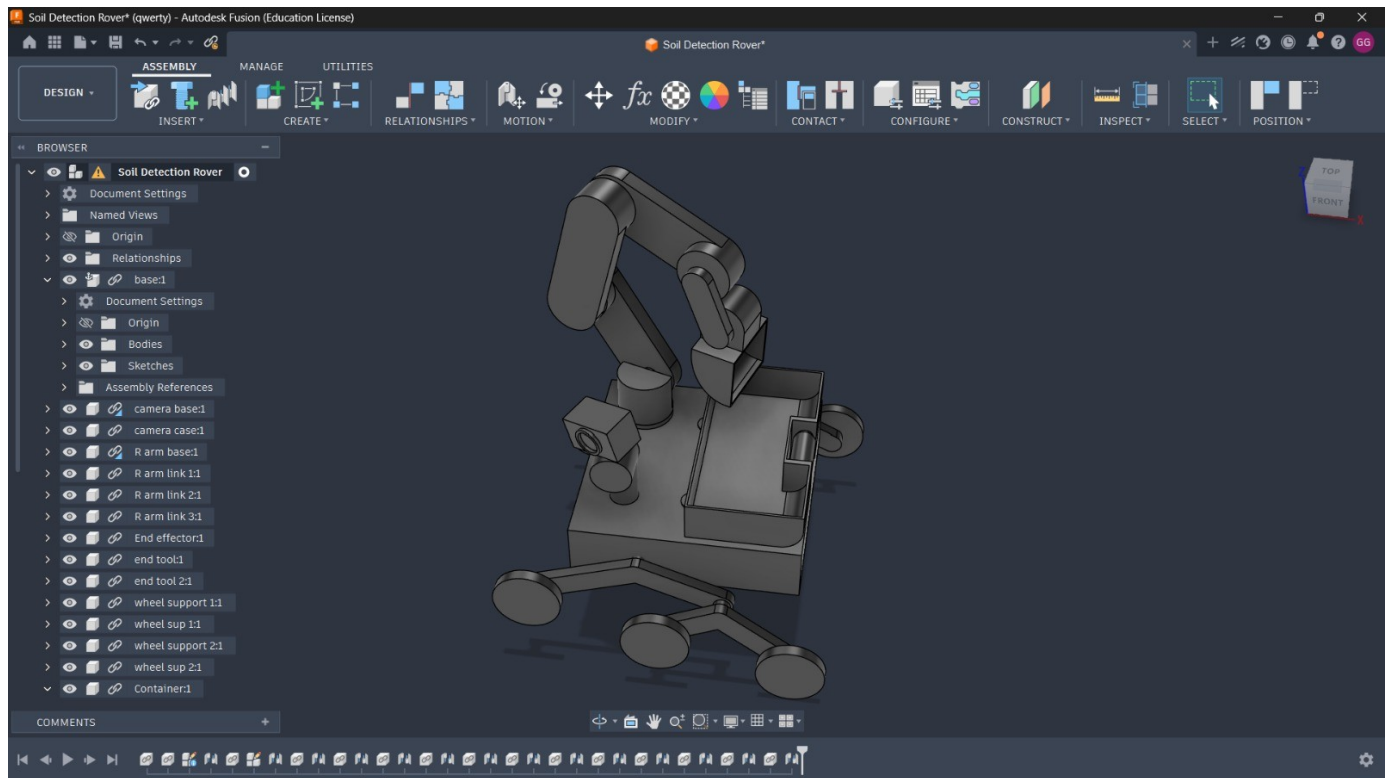
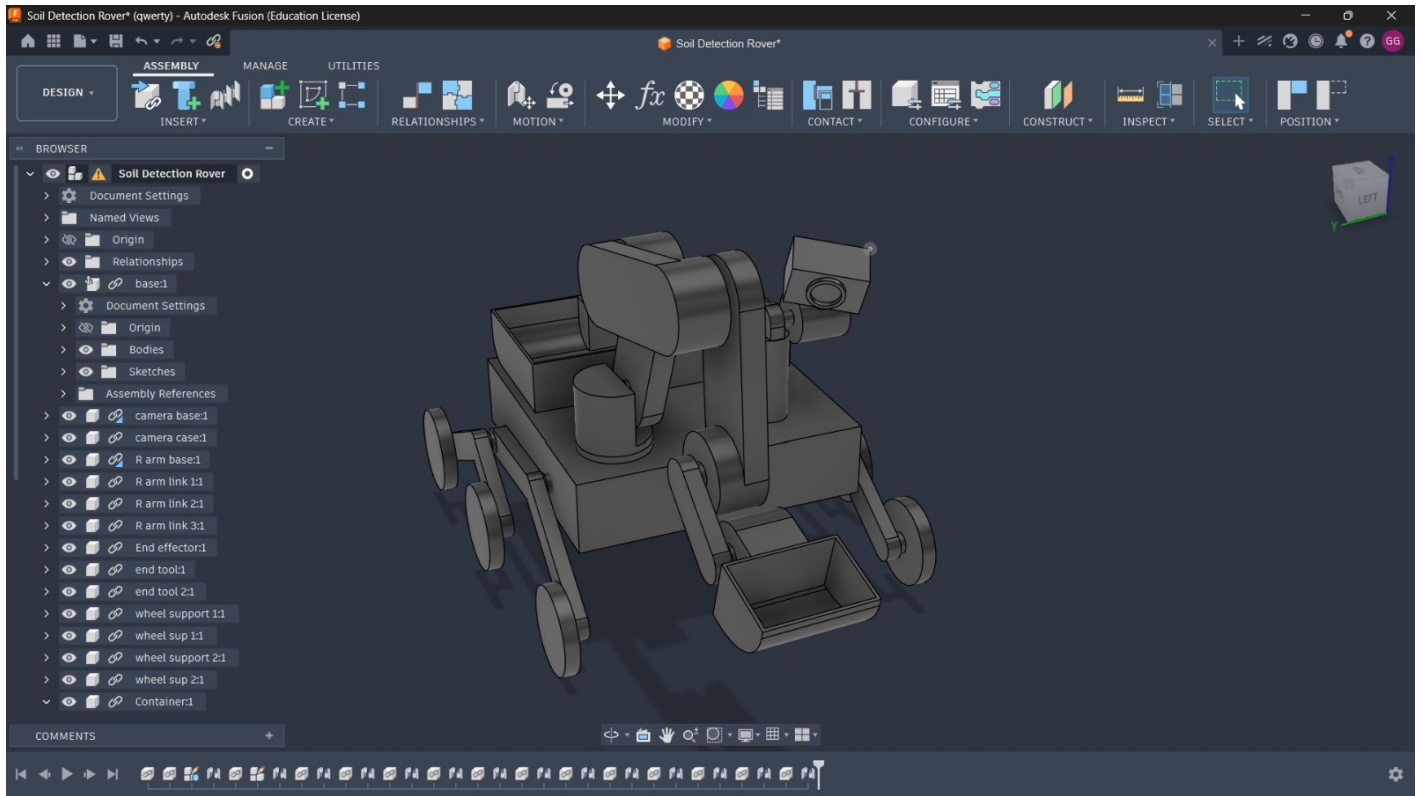
Dimension	Final CAD Value
Height	30 cm

9. Timeline for Robot Making with Milestones

Sr.	Activity / Milestone	No. of Days	Milestone / Output
1	Concept finalization	2	Functional architecture frozen
2	CAD design	4	Preliminary assembly and dimensions
3	Component selection & procurement	3	BOM and purchased parts
4	Chassis fabrication	3	Mobile base assembled
5	Arm and collector fabrication	5	Working collection mechanism
6	Electronics integration	3	Controller, drivers and sensors connected
7	Autonomous software	5	Detection, navigation and arm sequence
8	Closed-loop verification	4	Verify / re-collect logic tested
9	Integration and testing	4	Complete PoC demonstration



10. Proposed Outline Photograph



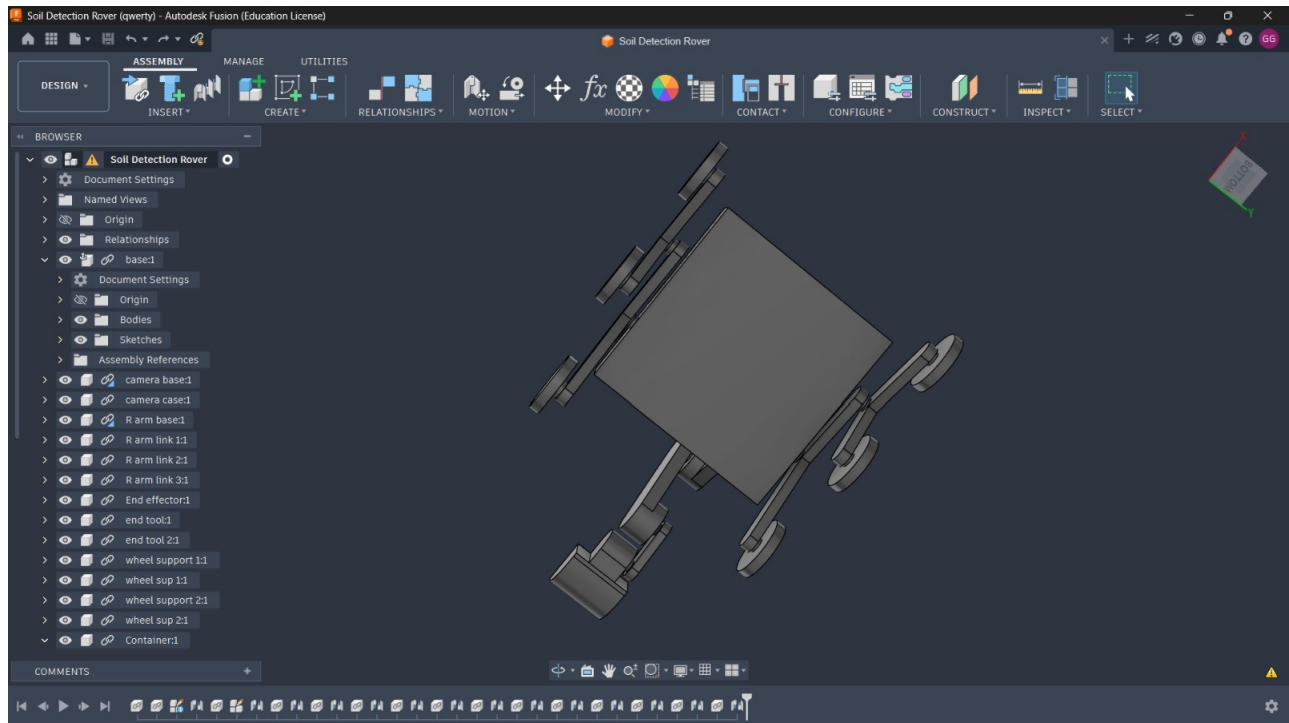
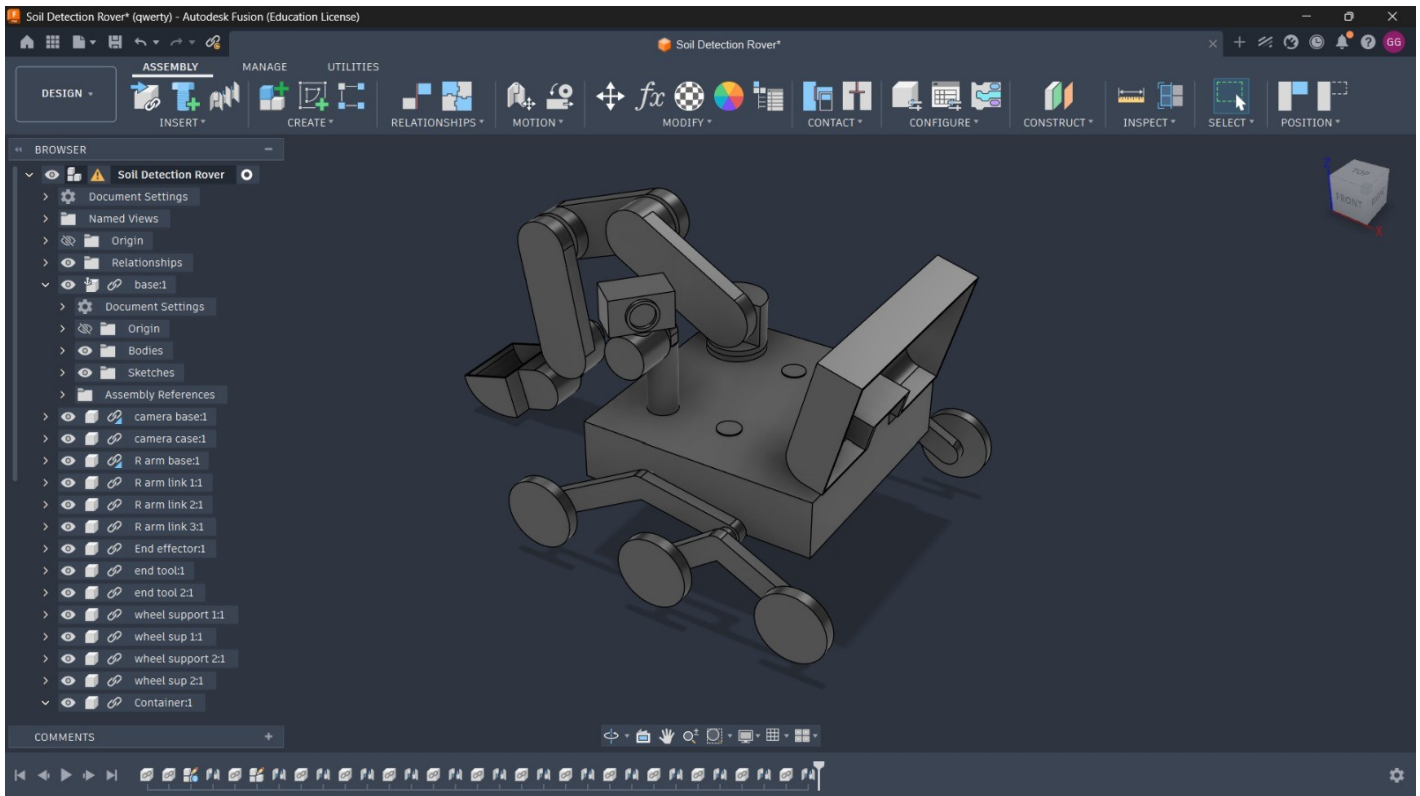


Figure 5. Proposed robot outline / concept render for evaluation understanding.